WEEK 4 - MULTI-PROCESS SIMULATOR TESTING

STC89C52 Educational Microcontroller Simulator

Testing Objective

Verify the Week 4 multi-process simulator, including independent UI, Core and Logging processes, inter-process communication, CPU execution, stack and FIFO operations, error detection, and the Fetch-Decode-Execute execution trace.

Test 1 - Three Independent Processes

Purpose: Verify that the UI, Core and Logger run as separate processes.

Method / Test Program:

Start the simulator and check the running Java processes.

Expected Result: Three independent Java processes should run: UI Process, Core Process and Logging Process.

Result: PASS

Test 2 - UI to Core Communication

Purpose: Verify communication between the UI Process and Core Process.

Method / Test Program:

```
MOV A,#10
MOV B,#20
ADD A,B
HALT
```

Expected Result: The program is sent from the UI to the Core and executed. Final expected values: A = 1EH and B = 14H.

Result: PASS

Test 3 - Core to Logger Communication

Purpose: Verify that execution information is sent from the Core Process to the Logging Process.

Method / Test Program:

Execute a loaded program using STEP or RUN and observe the Logger output.

Expected Result: The Logger receives execution events such as LOAD, RESET, STEP and HALT.

Result: PASS

Test 4 - STEP Operation

Purpose: Verify that STEP executes one instruction at a time.

Method / Test Program:

Load a valid program and press STEP repeatedly.

Expected Result: Exactly one instruction executes per STEP and the CPU state updates after each instruction.

Result: PASS

Test 5 - RUN Operation

Purpose: Verify that RUN automatically executes the complete program.

Method / Test Program:

Load a valid program containing HALT and press RUN once.

Expected Result: Instructions execute continuously until HALT is reached and the simulator terminates execution.

Result: PASS

Test 6 - STOP Operation

Purpose: Verify that execution can be stopped while RUN is active.

Method / Test Program:

Start RUN and press STOP before the program finishes.

Expected Result: Execution stops and the simulator displays the STOPPED state.

Result: PASS

Test 7 - Stack Operations

Purpose: Verify PUSH and POP operations and LIFO stack behavior.

Method / Test Program:

```
PUSH #15
PUSH #28
PUSH #42
POP A
POP A
POP A
HALT
```

Expected Result: Values are removed in LIFO order and SP returns to 07H after all pushed values are popped.

Result: PASS

Test 8 - FIFO Queue

Purpose: Verify ENQUEUE and DEQUEUE operations and FIFO behavior.

Method / Test Program:

```
ENQUEUE #10
ENQUEUE #20
ENQUEUE #30
DEQUEUE
DEQUEUE
DEQUEUE
HALT
```

Expected Result: Values leave the queue in the same order in which they were inserted.

Result: PASS

Test 9 - Invalid Instruction Detection

Purpose: Verify that invalid and unsupported instructions are detected by the UI error checker.

Method / Test Program:

```
MOV A,#10
ADD A,#5
MULT A,#2
SUB A
HELLO
HALT
```

Expected Result: Invalid or unsupported instructions are displayed in the Error Panel and execution is prevented when errors are present.

Result: PASS

Test 10 - Fetch, Decode and Execute Trace

Purpose: Verify that the simulator displays the CPU execution cycle.

Method / Test Program:

Execute instructions using STEP or RUN and observe the Execution Trace panel.

Expected Result: FETCH displays the fetched instruction, DECODE displays the opcode/operands, and EXECUTE displays the resulting CPU state change.

Result: PASS